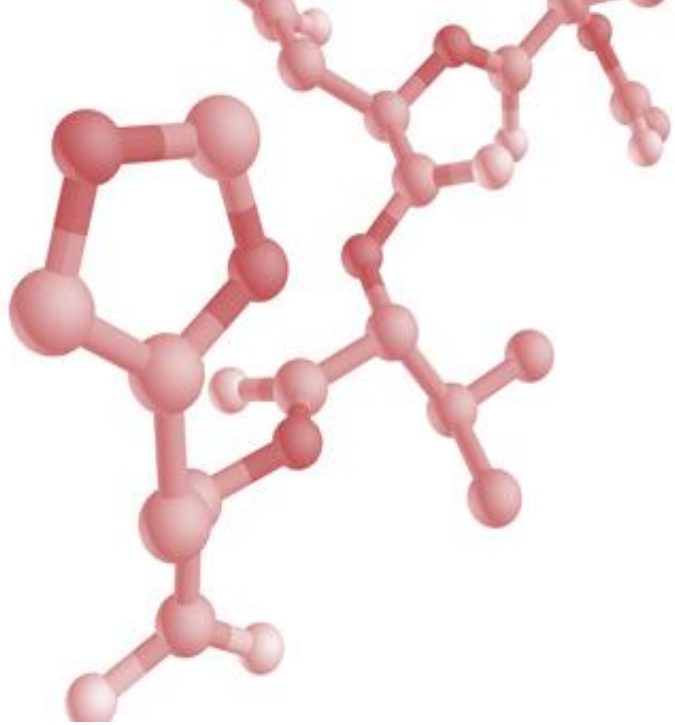
Dermican™

Revolumizing effect

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- **A pure tetrapeptide**
- **A targeted objective**: optimal **collagen fibers** functionality
- **Via a specific anchoring small proteoglycan**: **lumican**
- **For a firm and plump skin**

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In brief

Boosting collagen synthesis only is not sufficient to claim a thorough anti-aging efficacy.

Dermican™ is a step forward in the fight against skin aging, thanks to an original activity pathway.

Targets the metabolism of a specific proteoglycan of dermis Extra-Cellular Matrix (ECM): lumican.

For functional collagen fibers able to ensure good skin mechanical properties.

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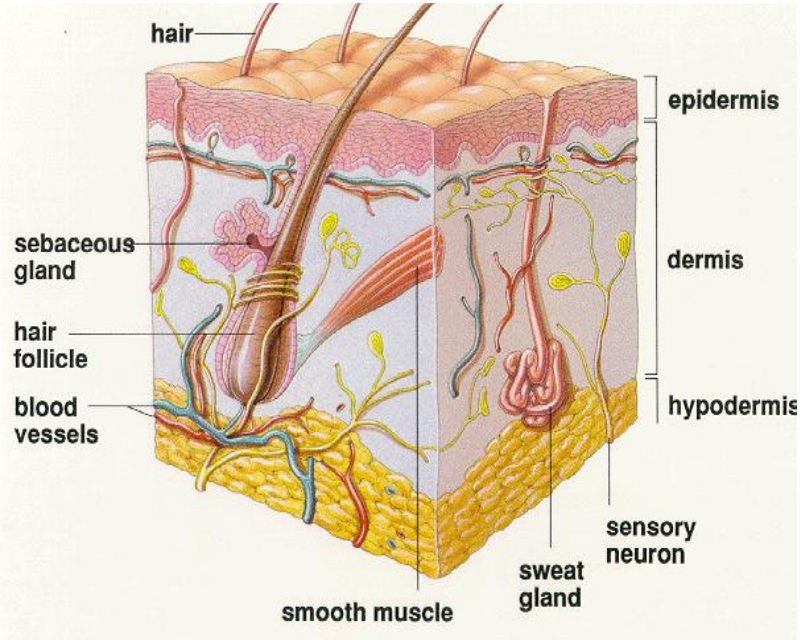
Background

Skin structure

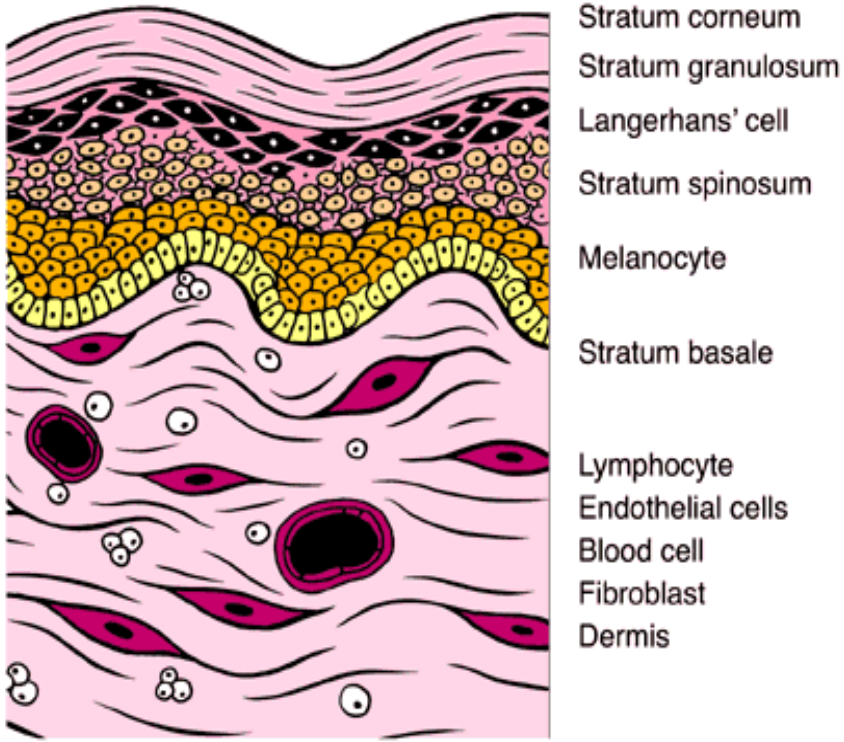
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Skin structure – A short reminder



www.agen.ufl.edu/.../lect/lect_19/lect_19.htm.



www.healthandage.com/html/res/primer/skin.htm.

Skin: made of 3 main layers: epidermis, dermis and hypodermis.

Dermis: supporting tissue of skin, responsible for skin mechanical properties: resistance, suppleness and elasticity.

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Extra-Cellular Matrix (ECM)

Largest component of normal skin.

Gel matrix surrounding cells (fibroblasts).

Composed of a variety of polysaccharides, water and proteins which give the skin remarkable properties.

Skin **elasticity**, **compressibility** and **strength** are influenced by molecules from the ECM :

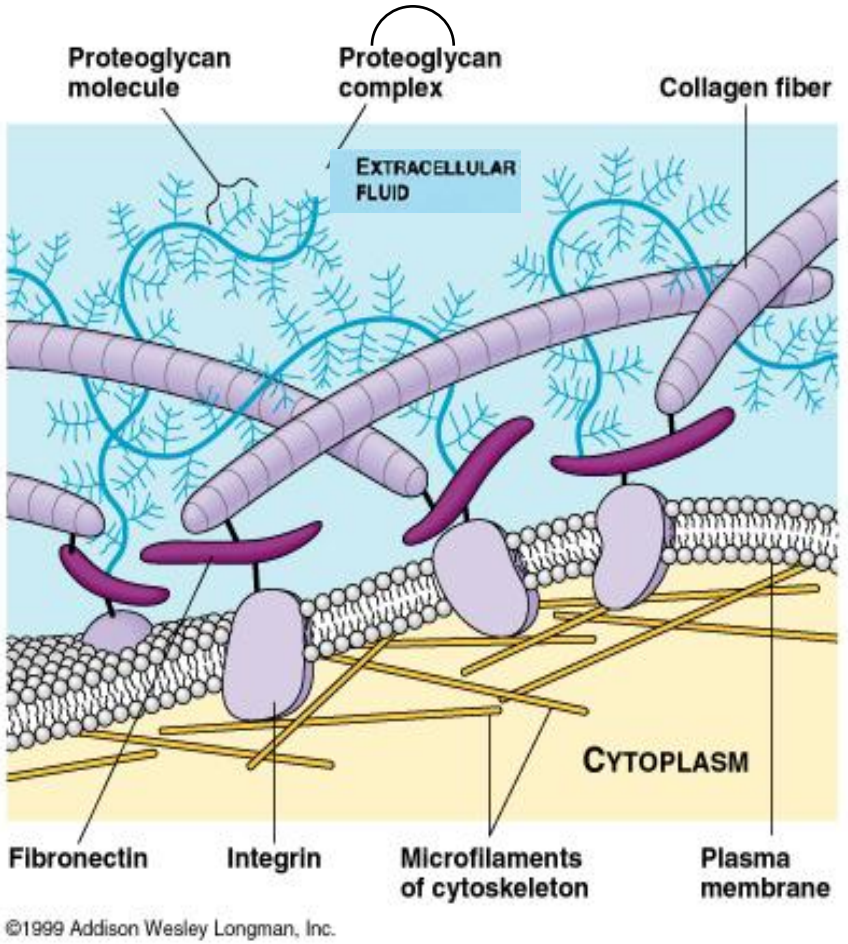
Structural proteins: collagen, elastin,

Proteoglycans: consist of glycosaminoglycan chains fixed to a linear core protein.

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Extra-Cellular Matrix (ECM)

Fibers of the Extra-Cellular Matrix (ECM)



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The Structure of Collagen

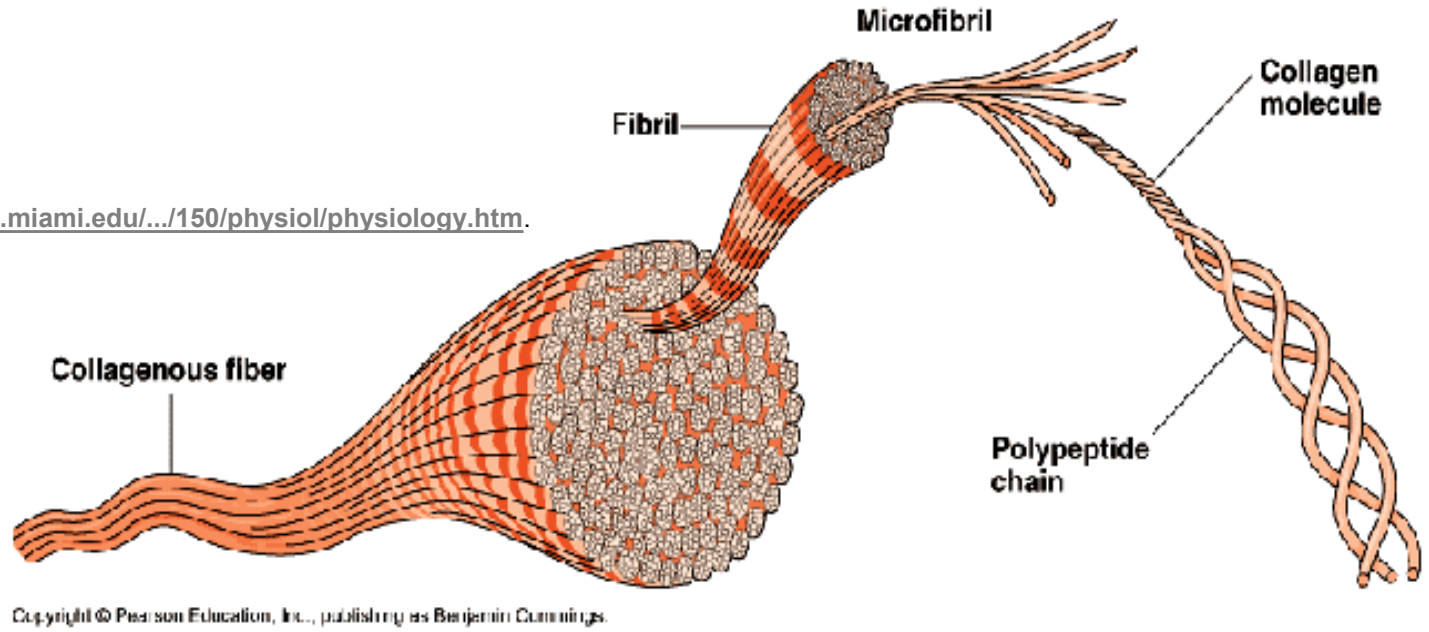


fig.cox.miami.edu/.../150/physiol/physiology.htm.

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- Collagen **fiber** = bundle of many macrofibrils.
- **Macrofibrils** = bundle of numerous microfibrils.
- **Microfibril** = composed of many tropocollagen helices.
- Tropocollagen helices = three polypeptide chains twisted together.

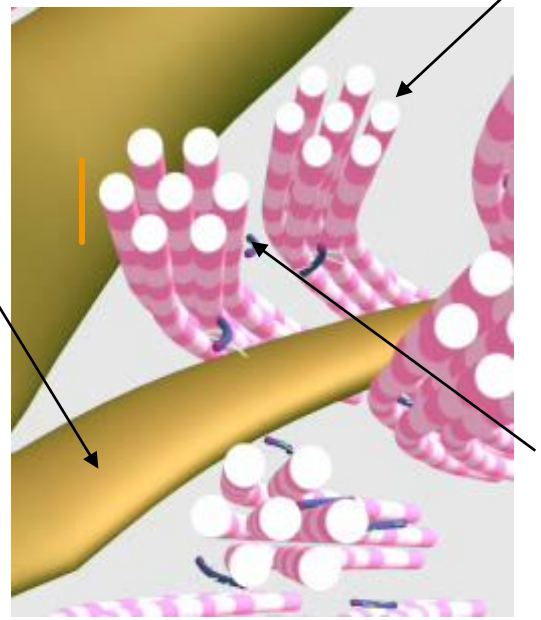
A specific **proteoglycan**, **lumican**, is involve in proper collagen structure.

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Lumican:
Role In Skin Integrity

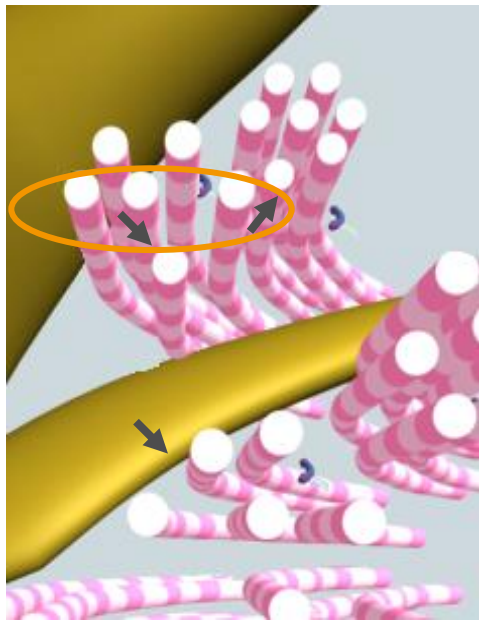
Lumican is involved in collagen I fibrils organization around dermal fibroblasts.

Young skin



Abundance of lumican
⇨ collagen I fibrils grouped together

Aged skin



Decrease of lumican
⇨ collagen I fibrils are more scattered

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Expression of Small Proteoglycans During Skin Aging

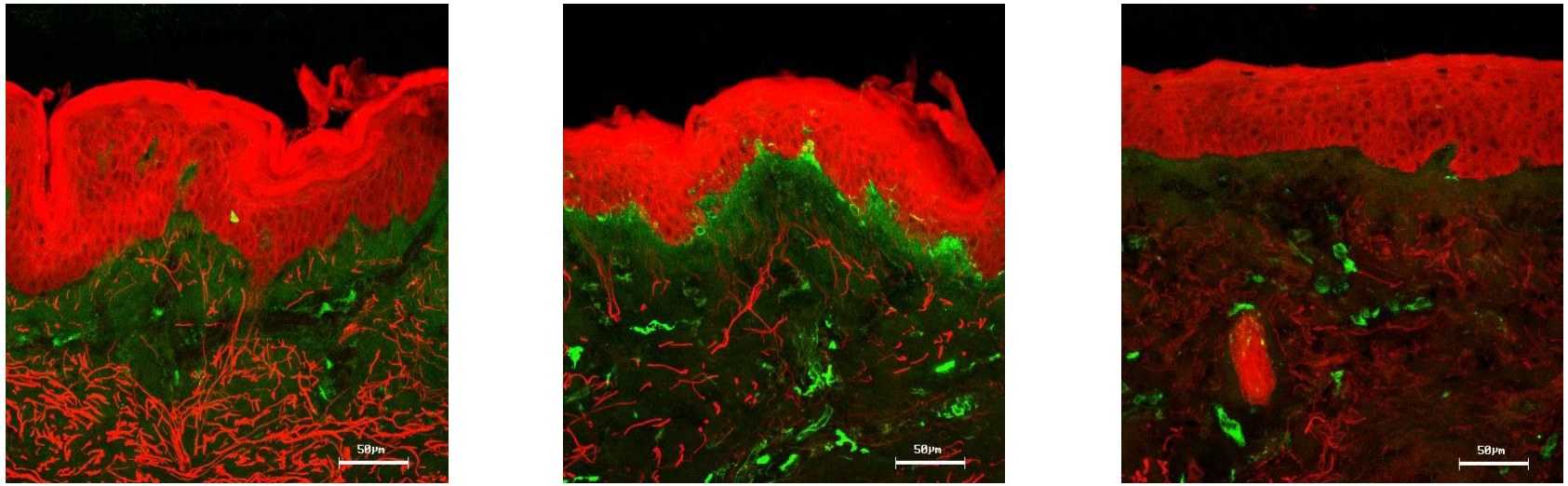
A fundamental study

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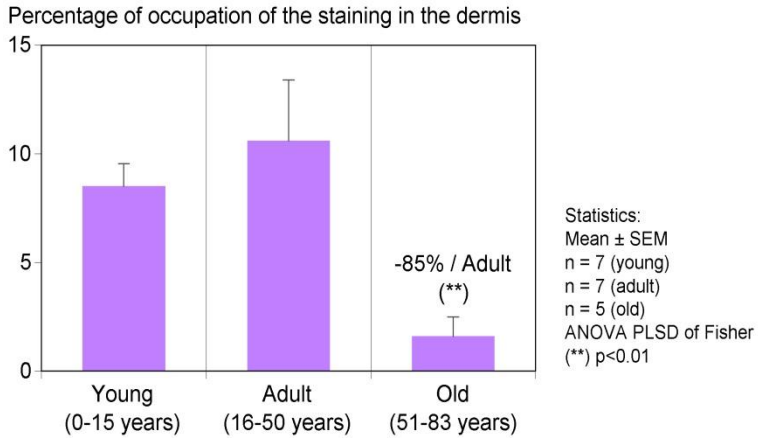
Care Creations™

Small proteoglycans in skin aging
Fundamental study

Evolution of lumican in dermis with aging (IHC and Confocal microscopy)



Lumican (in green on the pictures) **shows a clear decrease** (- 85%) **with age in dermis**



Statistics:
Mean ± SEM
n = 7 (young)
n = 7 (adult)
n = 5 (old)
ANOVA PLSD of Fisher
(**) p<0.01

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Conclusion

Lumican synthesis **decreases** with age.

The proper anchoring of collagen fibrils is no longer guaranteed, resulting in weakened extra cellular matrix, and a loss of **skin firmness**.

Our solution: Dermican™

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Dermican™
Composition

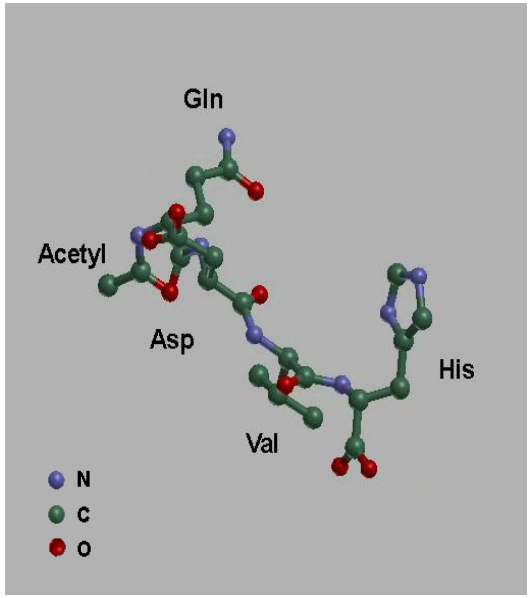
Dermican™ LS 9837: liquid version containing 1150 ppm of Acetyl Tetrapeptide-9

Composition: N-Acetyl-L-Glutaminyl-L-Aspartyl-L-Valyl-L-Histidine

Molecular weight: 539.5 g/mol

Other name: Acetyl QDVH

Structure:



Selected for its original anti-age biological action,

Particularly via lumican metabolism.

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Efficacy Tests

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Substantiation of Dermican™ concept

General biological profile of Tetrapeptide-9

DNA-Array test

Stimulation of genes coding for growth factors, structural proteins and structural proteoglycans

Effect on a small proteoglycan: lumican

Test on aged fibroblasts
Test on pseudo-dermis

Product tested : Acetyl Tetrapeptide-9

Effect on a structural protein: collagen

Stimulation of collagen I gene expression (qRT-PCR)
Stimulation of collagen I synthesis on cell culture

Product tested : Acetyl Tetrapeptide-9

Clinical study

In vivo test on 4 months, cream with 3% Dermican™

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Effect on a Small Proteoglycan:
Lumican – *Test On Aged Fibroblasts*

Protocol

Test performed on Acetyl Tetrapeptide-9

In vitro induced fibroblasts aging = several passages (>15)

↓

Seeding of human dermal fibroblasts

↓

Incubation for 2 days at 37°C, CO_2 = 5%

↓

Introduction of the products

↓

Control | IL-4 | Acetyl Tetrapeptide-9

↓

Incubation for 6 days at 37°C, CO_2 = 5%

↓

Visualization of lumican by immunocytochemistry

Quantification of the staining by image analysis

Hayflick's model: simulation of "aged fibroblasts" by multiple subcultures, before incubation with the product to be tested.

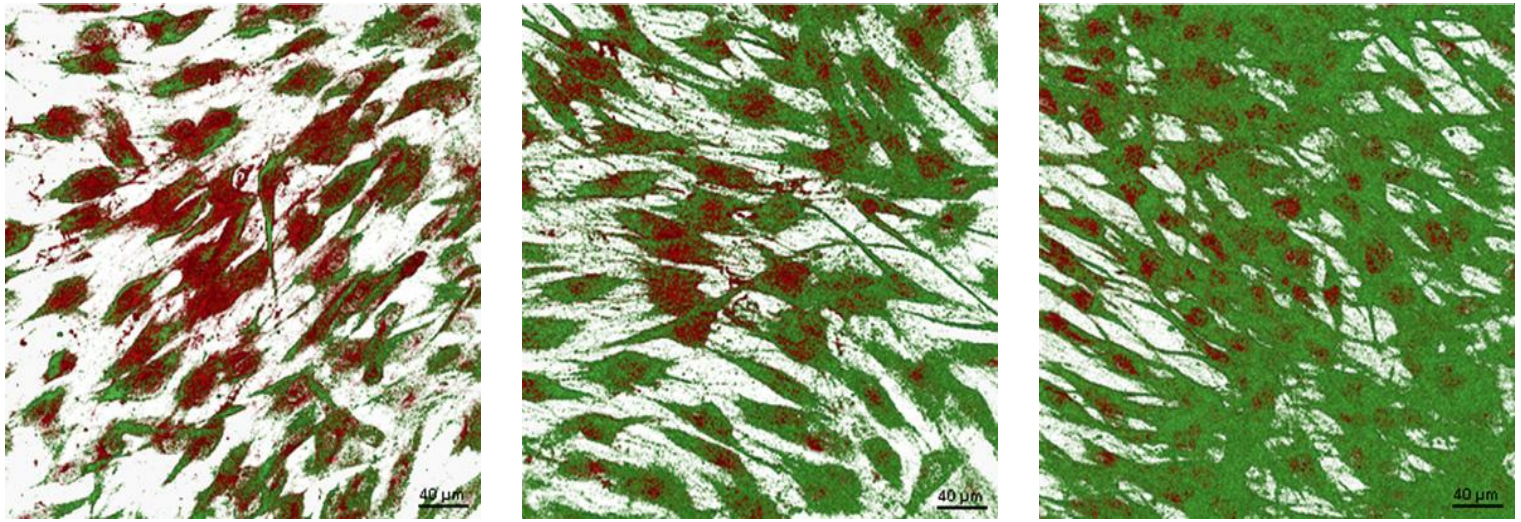
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Effect on a Small Proteoglycan:
Lumican – *Test On Aged Fibroblasts*

Results - Visualization

Test performed on Acetyl Tetrapeptide-9

Control | **IL-4 at 0.1 µg/ml** | **ACETYL TETRAPEPTIDE-9 at 2.2 µg/ml**



Lumican revealed in green

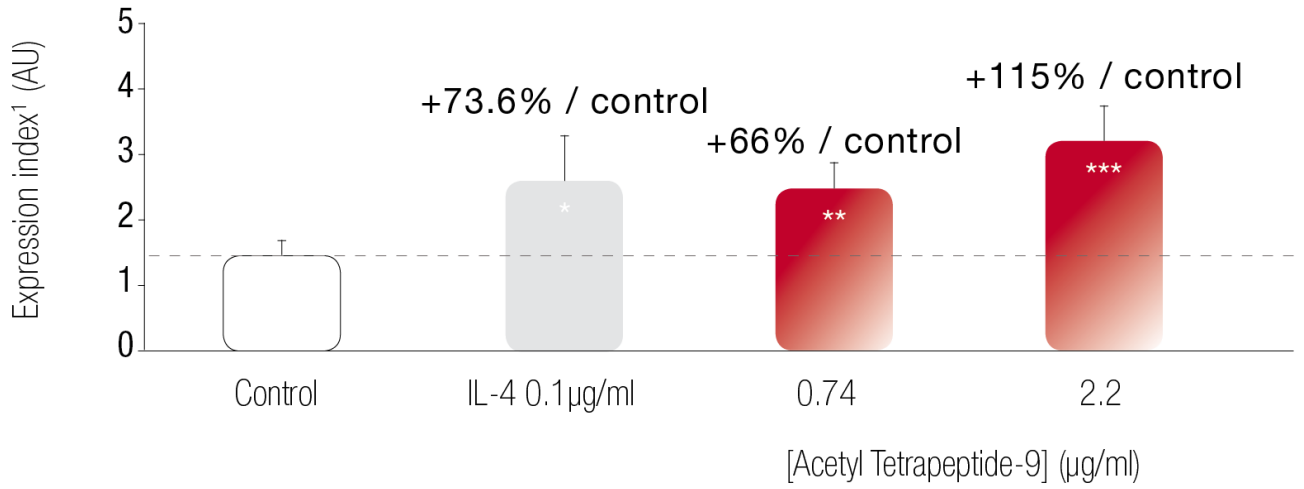
Interleukine-4 (IL-4) is used as a reference substance.

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Effect on a Small Proteoglycan:
Lumican – *Test On Aged Fibroblasts*

Results - Quantification

Test performed on Acetyl Tetrapeptide-9



Mean ± SEM
ANOVA PLSD of Fisher
n=12 (3 assays)
(*) p=0.02
(**) p<0.02
(***) p=0.0001

[1] Number of stained pixels x fluorescence intensity in green channel in arbitrary unit (AU)

Acetyl Tetrapeptide-9 has clearly increased **lumican synthesis** by aged fibroblasts, in a dose-dependent way.

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Effect On a Small Proteoglycan:
Lumican – *Test On Pseudo-dermis*

Protocol

Test performed on Acetyl Tetrapeptide-9

Collagen I + human fibroblasts + product

↓

Control Acetyl Tetrapeptide-9

↓

Collagen lattices

↓

Incubation for 6 days at 37°C, CO_2 = 5%

↓

Biopsies of the collagen lattices

↓

Visualization of lumican by immunohistochemistry

Quantification of the staining by image analysis

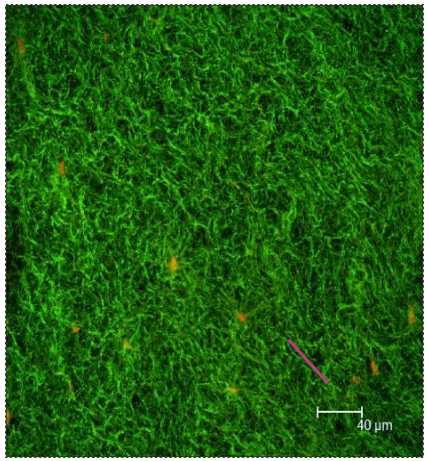
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Effect On a Small Proteoglycan:
Lumican – *Test On Pseudo-dermis*

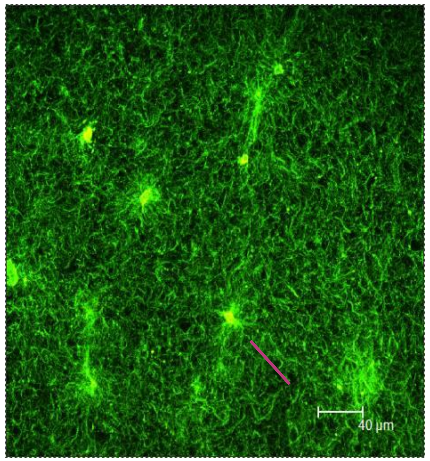
Results - Visualization

Test performed on Acetyl Tetrapeptide-9

Control



ACETYL TETRAPEPTIDE-9 at 2.2 µg/ml



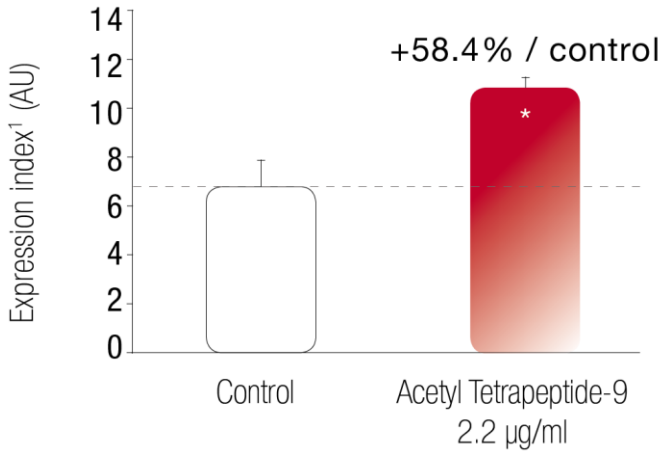
Lumican revealed in green

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Effect On a Small Proteoglycan:
Lumican – *Test On Pseudo-dermis*

Results - Quantification

Test performed on Acetyl Tetrapeptide-9



Mean ± SEM
on 3 assays
ANOVA PLSD of Fisher
(*) p<0.02

[1] Number of stained pixels x fluorescence intensity in green channel in arbitrary unit (AU)

Acetyl Tetrapeptide-9 has **clearly increased lumican synthesis** (+ 58%) vs control.

This confirms the first pillar of Dermican™ concept: booster effect of an anchoring proteoglycan, lumican.

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Effect On A Structural Protein:
Collagen – *Test On Fibroblasts*

Protocol

Stimulation of the expression of collagen I gene (*in vitro* test)

qRT-PCR test

Culture of human fibroblasts

Test performed on Acetyl Tetrapeptide-9

Seeding of human dermal fibroblasts

↓

Incubation for 3 to 5 days at 37°C, CO_2 = 5%

↓

Exchange of cell culture medium to a medium containing products

↓

Control TGF-β1 Acetyl Tetrapeptide-9

↓

Incubation for 24 hours at 37°C, CO_2 = 5%

↓

Extraction of total RNAs

↓

qRT-PCR reaction for COL1A1 gene

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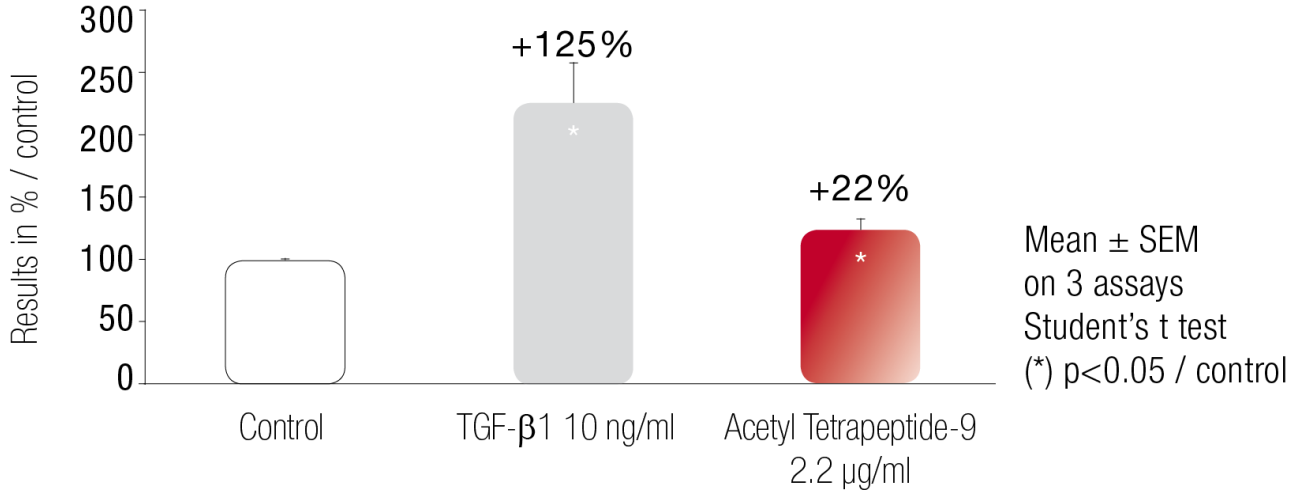
Effect On A Structural Protein:
Collagen – *Test On Fibroblasts*

Results - Quantification

Stimulation of the expression of collagen I gene (*in vitro* test)

qRT-PCR test

Test performed on Acetyl Tetrapeptide-9



The gene coding for collagen I in fibroblasts has been clearly boosted thanks to Acetyl Tetrapeptide-9 treatment. The final result, a **higher production of collagen I**, has been shown in the next test.

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Effect On A Structural Protein:
Collagen – *Test On Fibroblasts*

Protocol

Measurement of collagen I synthesis by fibroblasts
DOT-BLOT Technique

Test performed on Acetyl Tetrapeptide-9

Seeding of human dermal fibroblasts

↓

Incubation for 3 days at 37°C, CO_2 = 5%

↓

Exchange of cell culture medium to a medium containing products

↓

Control TGF-β1 Acetyl Tetrapeptide-9

↓

Incubation for 4 days at 37°C, CO_2 = 5%

↓

Extraction of cellular proteins:
1. measure of total proteins
2. measure of collagen I

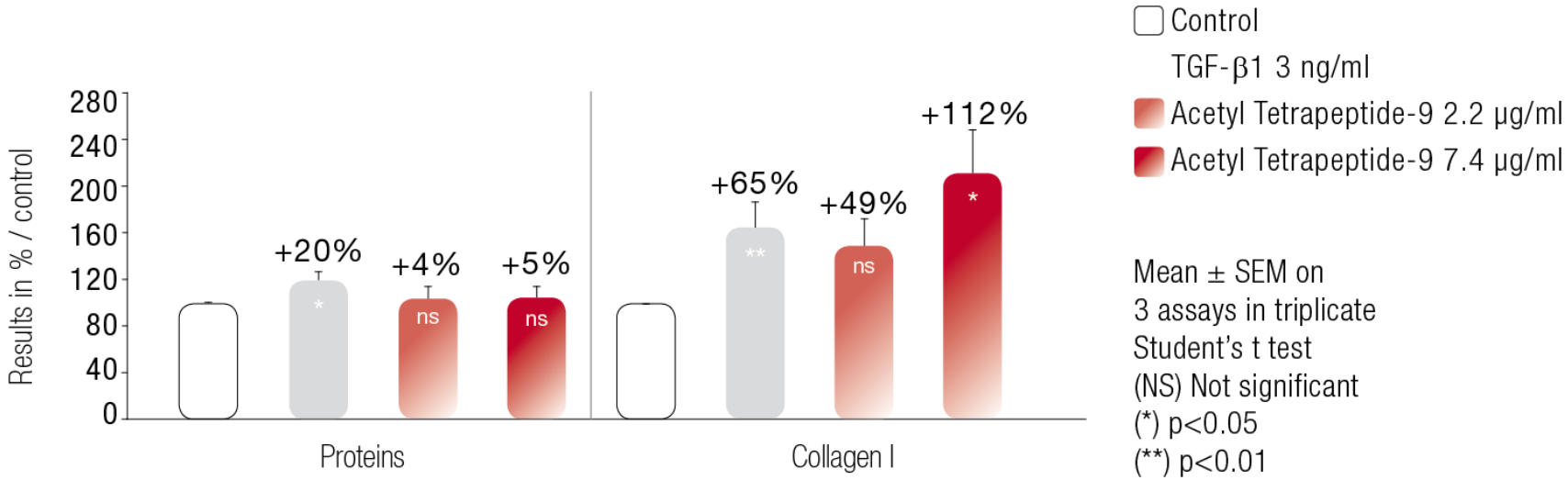
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Effect On A Structural Protein:
Collagen – *Test On Fibroblasts*

Results - Quantification

Measurement of collagen I synthesis by fibroblasts
DOT-BLOT Technique

Test performed on Acetyl Tetrapeptide-9



In presence of **Acetyl Tetrapeptide-9**, **production of collagen I by fibroblasts is clearly boosted**.
This confirms the second pillar of Dermican™ concept: booster effect of collagen I synthesis.

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Anti-aging effect – Clinical test

Protocol

Test performed on Dermican™

17 female volunteers, 45 to 55 years old,
with a loss of elasticity of the forearm

Bi-daily randomized application on the forearm for 16 weeks

Placebo cream



Cream with 0.9%
Dermican™

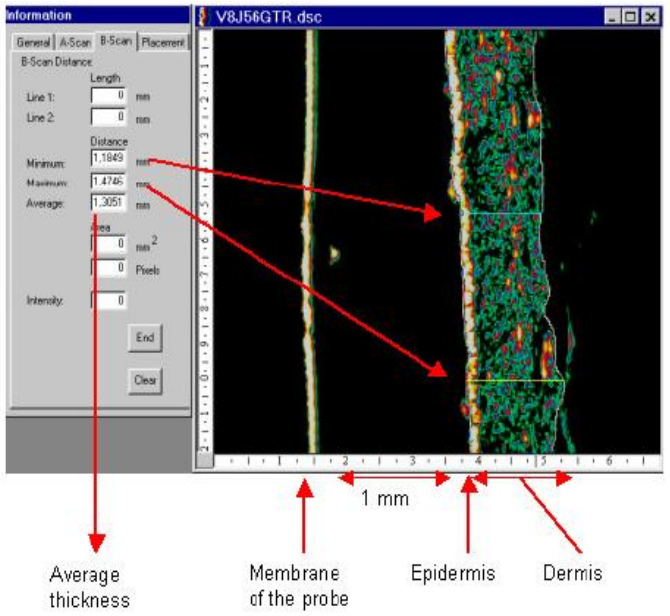
Measurement of the skin thickness by echography
Measurement of the skin firmness by cutometry

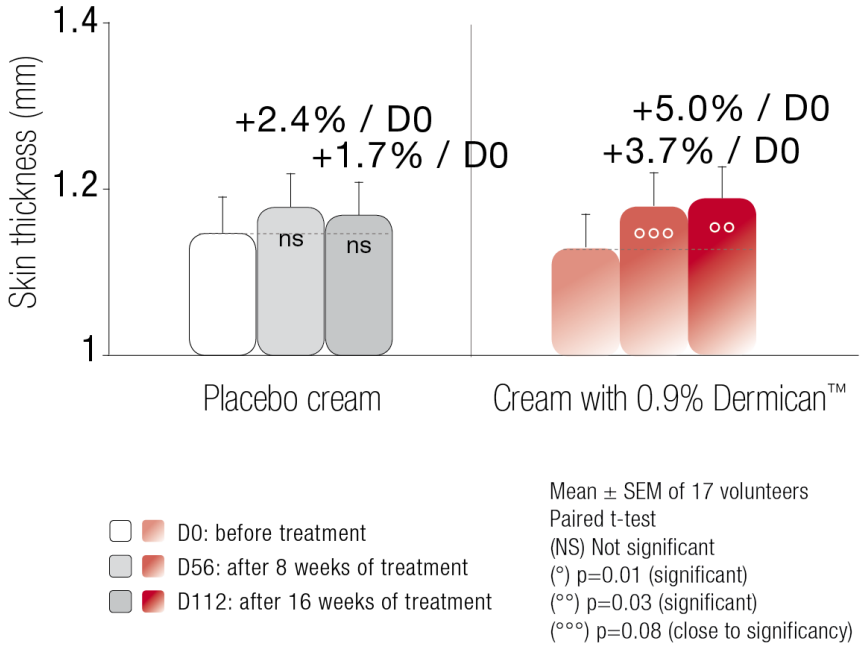
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Anti-aging effect – Clinical test

Results

Measurement of skin thickness by echography





Echography / ultrasound skin thickness evaluation using 2D DermaScan C equipment with a 20 MHz probe.

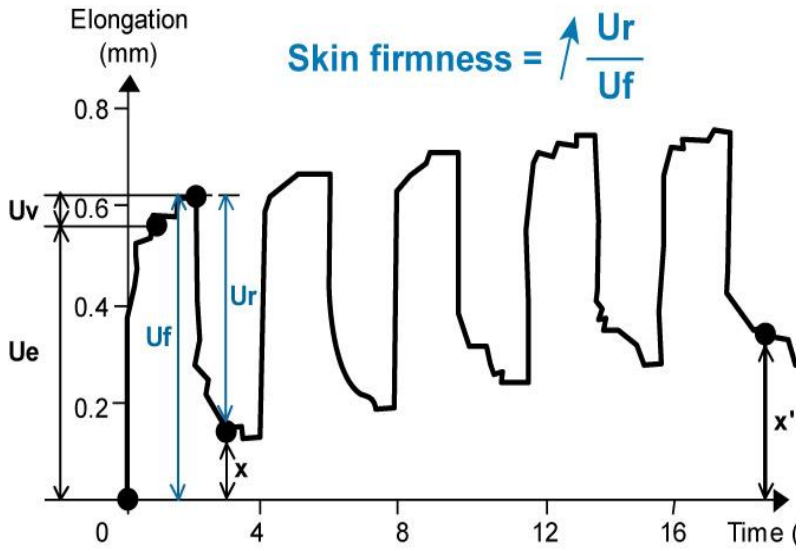
Significant increase of skin thickness after 8 weeks of treatment.
Effect confirmed after 16 weeks and positive for 67% of the volunteers.

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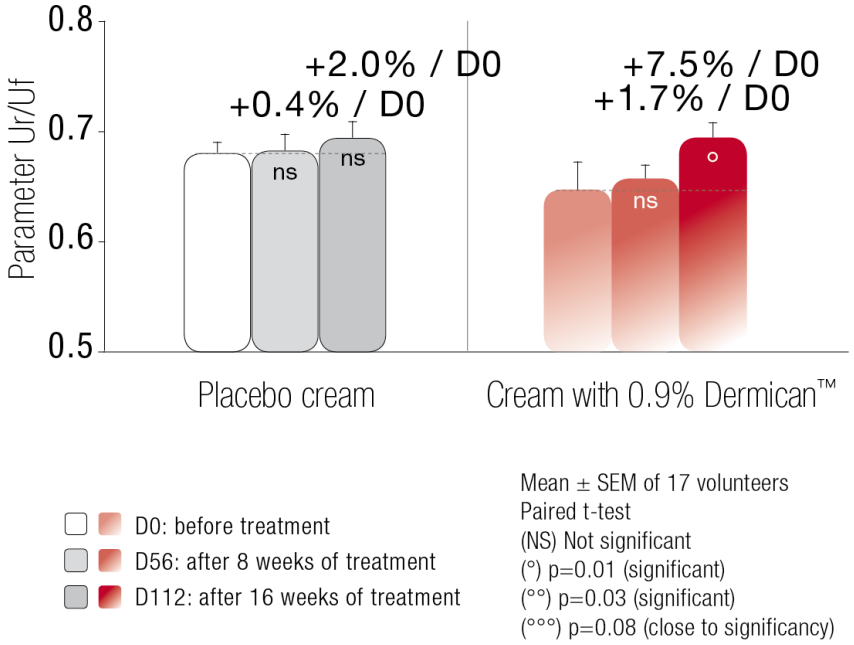
Anti-aging effect – Clinical test

Results

Measurement of skin firmness using Cutometer



Skin firmness evaluation using a SEM 575 CUTOMETER®.



Significant increase of skin firmness after 8 weeks of treatment.

Effect confirmed after 16 weeks and positive for 82% of the volunteers.

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Anti-aging effect – Clinical test

Conclusion

After 4 months of a bi-daily application of a cream with 0.9 % Dermican™:

67% of the volunteers showed a **positive effect** on **skin thickness**,

82% of the volunteers showed a **positive effect** on **skin firmness.**

Dermican™ clearly shows good **anti-aging effect**, acting both on skin structure and mechanical properties.
This effect is **perceivable** by the **end-users**.

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Cosmetic Properties

Dermcian™:

- A total and concerted action by targeting lumican and collagen I,
- Acts for a dense, regular, and functional network of collagen fibers,
- Improves skin firmness and structure,
- A well identified and efficient active, with detailed and substantiated mechanism of action.

Dermican™ perfectly fits to the cosmeceutical trend requesting well identified and efficient products, with detailed and substantiated mechanism of action

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Cosmetic Applications

- Anti-age face care,
- Firming body care,
- Anti-stretchmark care.

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Technical Data

Dermican™ LS 9837

Aspect	:	Colorless, clear liquid with a weak odor
Dose of use	:	0.3 - 1%
Solubility	:	Soluble in water, insoluble in oils and fats
INCI Name	:	Glycerin (and) Water (and) Acetyl Tetrapeptide-9
Preservatives	:	none

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